TPM3 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes TPM3 gene fusions from the msk_impact_50k_2026 study, covering 14 fusion events. 11/11 fusions with known frame status (100.0%) were in-frame. 9/14 fusions (64.3%) retained the Tropomyosin domain. Domain retention was tested with Fisher's exact test (p=1, not statistically significant at alpha=0.05). No literature benchmark is configured for TPM3.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 14 analyzed fusion events, identifying 2 distinct fusion partner genes. The most frequent partner was NTRK1, observed in 12 events.

Domain retention

Tropomyosin domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 6/11 (54.5%) of in-frame fusions retained the domain, p=1 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 1.99998e-05.

Domain disruption

Domain-disruption analysis was skipped: TPM3 has no disruption_required_domains configured; domain-disruption analysis was skipped.

Junction concentration

Junction concentration could not be tested for this run: 4 event(s) with a known junction exon inside the transcript; at least 5 are needed

Cutpoint detection

Cutpoint detection scanned 11 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 259 aa (exon 8/9 boundary; label-permutation p=0.003996). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is -25 aa from the nearest configured domain boundary at 284 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=8) and "not_retained" (n=3). For Frame_status, retained: 6/6 (100.0%, 95% CI 61.0%-100.0%); not_retained: 3/3 (100.0%, 95% CI 43.9%-100.0%). Welch's t-test comparing Tumor_variant_count between groups gave means of 26.75 vs 62.6667, p=0.0927984 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for TPM3 in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (9 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 2 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: TPM3 has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: TPM3 has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (7 row(s) in window_scan, 2 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
NTRK1	12	0.8571428571428571	0.46989743472591233	None	0.3374485596707819	0.5240843401713946	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.5970536672401149	1
ROS1	2	0.14285714285714285	0.46989743472591233	None	0.0	0.5240843401713946	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.2659249194447005	2

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
259	0	8	3	0	0.0	0.006060606060606061	2.2174839442139063
285	1	7	3	0	0.0	0.024242424242424242	1.615423952885944

domain_retention tables

frame_domain_contingency_table

6	2
3	0

frame_domain_contingency_tables_by_tumor_type

stratum	table
COAD	[1, 1], [0, 0]
GB	[1, 0], [0, 0]
LUAD	[2, 0], [0, 0]
SARCNOS	[1, 0], [1, 0]
THPA	[1, 1], [2, 0]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	Tropomyosin	11	8	3	0	0.8940677966101694

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
2	11	11	1.0

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
NTRK1	12	12	12	9
ROS1	2	2	2	2

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
1	154142000	154143000	3	EVT-P-0021170-T02-IM6-4, EVT-P-0026375-T01-IM6-8, EVT-P-0046877-T02-IM6-10
1	154130000	154131000	1	EVT-P-0050767-T01-IM6-13
1	154133000	154134000	1	EVT-P-0028865-T02-IM6-2
1	154134000	154135000	1	EVT-P-0025815-T02-IM6-7
1	154135000	154136000	1	EVT-P-0003207-T01-IM5-1
1	154136000	154137000	1	EVT-P-0049078-T01-IM6-11
1	154138000	154139000	1	EVT-P-0014143-T01-IM5-3
1	154139000	154140000	1	EVT-P-0021467-T01-IM6-6
1	154141000	154142000	1	EVT-P-0069266-T01-IM7-14

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
286	7	7	8910	False

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
259	3	3	425	False

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
85	285	200	4	7	1	7	0.0	0.024242424242424242	1.615423952885944	EVT-P-0021170-T02-IM6-4, EVT-P-0026375-T01-IM6-8, EVT-P-0046877-T02-IM6-10, EVT-P-0069266-T01-IM7-14
185	285	100	4	7	1	7	0.0	0.024242424242424242	1.615423952885944	EVT-P-0021170-T02-IM6-4, EVT-P-0026375-T01-IM6-8, EVT-P-0046877-T02-IM6-10, EVT-P-0069266-T01-IM7-14
235	285	50	4	7	1	7	0.0	0.024242424242424242	1.615423952885944	EVT-P-0021170-T02-IM6-4, EVT-P-0026375-T01-IM6-8, EVT-P-0046877-T02-IM6-10, EVT-P-0069266-T01-IM7-14
286	311	25	7	4	7	1	None	0.024242424242424242	1.615423952885944	EVT-P-0003207-T01-IM5-1, EVT-P-0014143-T01-IM5-3, EVT-P-0021467-T01-IM6-6, EVT-P-0025815-T02-IM6-7, EVT-P-0028865-T02-IM6-2, ... (7 total)
286	336	50	7	4	7	1	None	0.024242424242424242	1.615423952885944	EVT-P-0003207-T01-IM5-1, EVT-P-0014143-T01-IM5-3, EVT-P-0021467-T01-IM6-6, EVT-P-0025815-T02-IM6-7, EVT-P-0028865-T02-IM6-2, ... (7 total)
286	386	100	7	4	7	1	None	0.024242424242424242	1.615423952885944	EVT-P-0003207-T01-IM5-1, EVT-P-0014143-T01-IM5-3, EVT-P-0021467-T01-IM6-6, EVT-P-0025815-T02-IM6-7, EVT-P-0028865-T02-IM6-2, ... (7 total)
286	486	200	7	4	7	1	None	0.024242424242424242	1.615423952885944	EVT-P-0003207-T01-IM5-1, EVT-P-0014143-T01-IM5-3, EVT-P-0021467-T01-IM6-6, EVT-P-0025815-T02-IM6-7, EVT-P-0028865-T02-IM6-2, ... (7 total)

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
286	311	25	7	4	7	1	None	0.024242424242424242	1.615423952885944	EVT-P-0003207-T01-IM5-1, EVT-P-0014143-T01-IM5-3, EVT-P-0021467-T01-IM6-6, EVT-P-0025815-T02-IM6-7, EVT-P-0028865-T02-IM6-2, ... (7 total)
235	285	50	4	7	1	7	0.0	0.024242424242424242	1.615423952885944	EVT-P-0021170-T02-IM6-4, EVT-P-0026375-T01-IM6-8, EVT-P-0046877-T02-IM6-10, EVT-P-0069266-T01-IM7-14

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0003207-T01-IM5-1	P-0003207-T01-IM5	P-0003207	Non-Small Cell Lung Cancer	LUAD	ROS1-TPM3	ROS1	in-frame	five_prime	154135010	10	286	True	286	True	exact match	retained	1.0	False	Tropomyosin			NA | Protein fusion: in frame (TPM3-ROS1)	NA
EVT-P-0028865-T02-IM6-2	P-0028865-T02-IM6	P-0028865	Non-Small Cell Lung Cancer	LUAD	ROS1-TPM3	ROS1	in-frame	five_prime	154133648	10	286	True	286	True	exact match	retained	1.0	False	Tropomyosin			TPM3 (NM_001278190) - ROS1 (NM_002944) fusion (TPM3 exons 1-6 with ROS1 exons 35-43): t(1;6)(q21.3;q22.1)(chr1:g.154133648::chr6:g.117643589) | Protein Fusion: in frame {TPM3:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014143-T01-IM5-3	P-0014143-T01-IM5	P-0014143	Colorectal Cancer	COAD	TPM3-NTRK1	NTRK1	in-frame	five_prime	154138845	10	286	True	286	True	exact match	retained	1.0	False	Tropomyosin			TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-10 fused in-frame to NTRK1 exons 9-17): c.*1568:TPM3_c.1178-173:NTRK1inv | Protein fusion: in frame (TPM3-NTRK1)	NA
EVT-P-0021170-T02-IM6-4	P-0021170-T02-IM6	P-0021170	Soft Tissue Sarcoma	SARCNOS	TPM3-NTRK1	NTRK1	in-frame	five_prime	154142641	8	259	True	258	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.894067796610 1694	True			Tropomyosin	TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-8 fused to NTRK1 exons 12-17): c.775+235:TPM3_c.1355-94:NTRK1inv | Protein Fusion: in frame {TPM3:NTRK1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021170-T03-IM6-5	P-0021170-T03-IM6	P-0021170	Soft Tissue Sarcoma	SARCNOS	TPM3-NTRK1	NTRK1	in-frame	five_prime	154142641	8	259	True	258	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.894067796610 1694	True			Tropomyosin	TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-8 fused to NTRK1 exons 12-17): c.775+235:TPM3_c.1355-94:NTRK1inv | Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-0021467-T01-IM6-6	P-0021467-T01-IM6	P-0021467	Glioma	GB	TPM3-NTRK1	NTRK1	in-frame	five_prime	154139761	10	286	True	286	True	exact match	retained	1.0	False	Tropomyosin			TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-6 fused in-frame to NTRK1 exons 9-17): c.*652:TPM3_c.1178-137:NTRK1inv | Protein Fusion: in frame {TPM3:NTRK1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025815-T02-IM6-7	P-0025815-T02-IM6	P-0025815	Soft Tissue Sarcoma	SARCNOS	TPM3-NTRK1	NTRK1	in-frame	five_prime	154134381	10	286	True	286	True	exact match	retained	1.0	False	Tropomyosin			TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-6 with NTRK1 exons 10-17): c.*6032:TPM3_c.1195+38:NTRK1inv | Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-0026375-T01-IM6-8	P-0026375-T01-IM6	P-0026375	Thyroid Cancer	THPA	TPM3-NTRK1	NTRK1	in-frame	five_prime	154142521	8	259	True	258	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.894067796610 1694	True			Tropomyosin	TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-8 fused to NTRK1 exons 9-17): c.775+355:TPM3_c.1178-129:NTRK1inv | Protein Fusion: in frame {TPM3:NTRK1}	NA

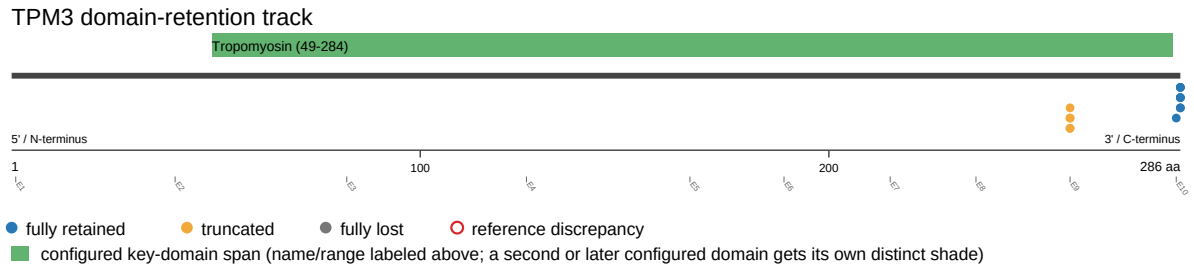
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0026375-T04-IM6-9	P-0026375-T04-IM6	P-0026375	Thyroid Cancer	THAP	TPM3-NTRK1	NTRK1	in-frame	five_prime	154142521	8	259	True	258	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.894067796610 1694	True			Tropomyosin	TPM3 (NM_152263) - NTRK1 (NM_002529) fusion: c.775+355:TPM3_c.1178-129:NTRK1inv | Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-0046877-T02-IM6-10	P-0046877-T02-IM6	P-0046877	Thyroid Cancer	THPA	TPM3-NTRK1	NTRK1	in-frame	five_prime	154142216	8	259	True	258	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.894067796610 1694	True			Tropomyosin	TPM3 (NM_152263) - NTRK1 (NM_002529) fusion: c.776-357:TPM3_c.1177+60:NTRK1inv | Protein Fusion: in frame {TPM3:NTRK1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0049078-T01-IM6-11	P-0049078-T01-IM6	P-0049078	Colorectal Cancer	COAD	TPM3-NTRK1	NTRK1	unknown	five_prime	154136376	10	286	True	286	True	exact match	retained	1.0	False	Tropomyosin			TPM3 (NM_001278190) - NTRK1 (NM_002529) fusion: c.395-6179:TPM3_c.963:NTRK1inv | Protein Fusion: mid-exon {TPM3:NTRK1}	NA
EVT-P-0049078-T04-IM6-12	P-0049078-T04-IM6	P-0049078	Colorectal Cancer	COAD	TPM3-NTRK1	NTRK1	unknown	five_prime	154136376	10	286	True	286	True	exact match	retained	1.0	False	Tropomyosin			TPM3 (NM_001278190) - NTRK1 (NM_002529) fusion: c.395-6179:TPM3_c.963:NTRK1inv | Protein Fusion: mid-exon {TPM3:NTRK1}	NA
EVT-P-0050767-T01-IM6-13	P-0050767-T01-IM6	P-0050767	Thyroid Cancer	THPA	TPM3-NTRK1	NTRK1	in-frame	five_prime	154130851	10	286	True	286	True	exact match	retained	1.0	False	Tropomyosin			TPM3(NM_153469) - NTRK1 (NM_002529) Fusion(TPM3 exon 12 with NTRK1 exon12): c.665-654:TPM3_c.1355-60:NTRK1inv | Protein Fusion: in frame {TPM3:NTRK1}	NA

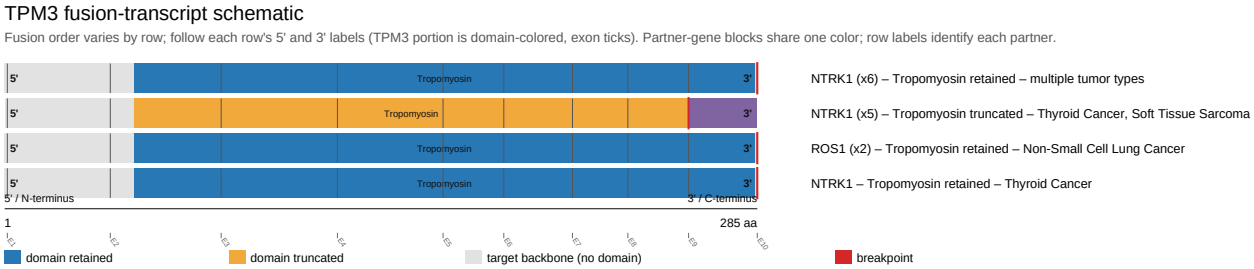
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0069266-T01-IM7-14	P-0069266-T01-IM7	P-0069266	Thyroid Cancer	THPA	TPM3-NTRK1	NTRK1	unknown	five_prime	154141499	9	285	True	284	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Tropomyosin			TPM3 (NM_152263) - NTRK1 (NM_002529) fusion: c.854+282:TPM3_c.1027:NTRK1inv | Protein Fusion: mid-exon {TPM3:NTRK1}	NA

Figures

domain retention outliers



fusion schematic



Partner-block colors are decorative; row labels identify partners.